Do syntax-like representations emerge in LLMs? Evidence from structural priming

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Large language models (LLMs) learn from unannotated text via next-token prediction, prompting the question of whether they nonetheless develop supra-token, syntax-like representations that support hierarchical sentence production. We investigated this question by preregistering and conducting seven structural priming replications in ChatGPT, sampling classic paradigms used to diagnose syntactic representations in human production and comprehension. Structural priming—speakers' tendency to reuse recently experienced structures—has long been taken as evidence for abstract structural frames in humans and provides a direct behavioral lens on whether LLMs exhibit analogous representations (Branigan & Pickering, 2017; Cai & Zhao, 2024).

METHODS. We treated ChatGPT as the "participant". Using MacBehaviour (Duan et al., 2024), each run presented one independent trial to avoid cross-trial carryover. Stimuli mirrored human experiments and targeted dative alternations, clause-position manipulations, garden-path ambiguity, VP ellipsis, plausibility-driven reanalysis, and cross-language transfer. Tasks included prime generation or reading, yes/no comprehension for attention control, and target sentence completion. We used API-based collection and executed 1000 runs per item version to ensure stable estimates. Prime and target outputs were automatically coded with a high-accuracy dependency-based algorithm (spaCy en_core_web_trf), validated against human coding in related corpora. Analyses used mixed-effects logistic regression over DO vs. PO (and transitive vs. intransitive) target outcomes, with maximal random effects justified by the data and convergence safeguards.

RESULTS. **Study 1** (morphology) showed robust dative priming with insensitivity to tense, but measurable increases in priming when aspect or number morphology matched across prime and target—departing from human insensitivity to morphology (Pickering & Branigan, 1998) and suggesting stronger form-level coupling in the model. **Study 2** (longevity) found that abstract priming persisted over 0-, 2-, and 6-lag separations; notably, a lexical boost due to verb overlap was reliable and showed little evidence of decay, contrasting with human findings that lexical boosts diminish over intervening material (Hartsuiker et al., 2008). **Study 3** (clause position) demonstrated comparable priming whether the prime/target structure appeared in main or subordinate clauses, aligning with human generalization across clause positions (Branigan et al., 2006). **Study 4** (VP ellipsis) yielded no difference between ellipsis and full-form primes relative to a baseline; combined with strong contextual memory in LLMs, this pattern is more consistent with an inability to discriminate one- versus two-structure exposure than with clear syntactic reconstruction of elided VPs (Cai et al., 2013). **Study 5** (implausibility) replicated a reduction of priming after implausible primes and showed modulation by interpretation: when model answers suggested nonliteral reinterpretation, priming differed from trials with literal interpretations, broadly paralleling human evidence for structural reanalysis under plausibility constraints (Cai et al., 2022). **Study 6** (garden-path) revealed lingering influence of abandoned transitive parses: ambiguous primes increased transitive completions relative to unambiguous controls, and the effect was further shaped by NP length, plausibility, and verb repetition—mirroring human demonstrations that initial, inappropriate analyses remain partially active (van Gompel et al., 2006). Finally, **Study 7** (cross-language priming) showed cross-language priming with a translation-equivalent boost that was smaller than the within-language lexical boost, recapitulating bilingual patterns (Schoonbaert et al., 2007).

DISCUSSION. These results indicate that ChatGPT reuses abstract structural frames in ways that substantially overlap with human structural priming signatures while also

exhibiting model-specific properties: stronger or more persistent coupling to lexical/morphological overlap, and ambiguous evidence for ellipsis reconstruction when contextual exposure is hard to equate. The convergences suggest emergent, supra-token representations supporting hierarchical combination; the divergences constrain how such representations are organized and accessed in LLMs, refining theoretical claims about the representational status of syntax under next-token learning.

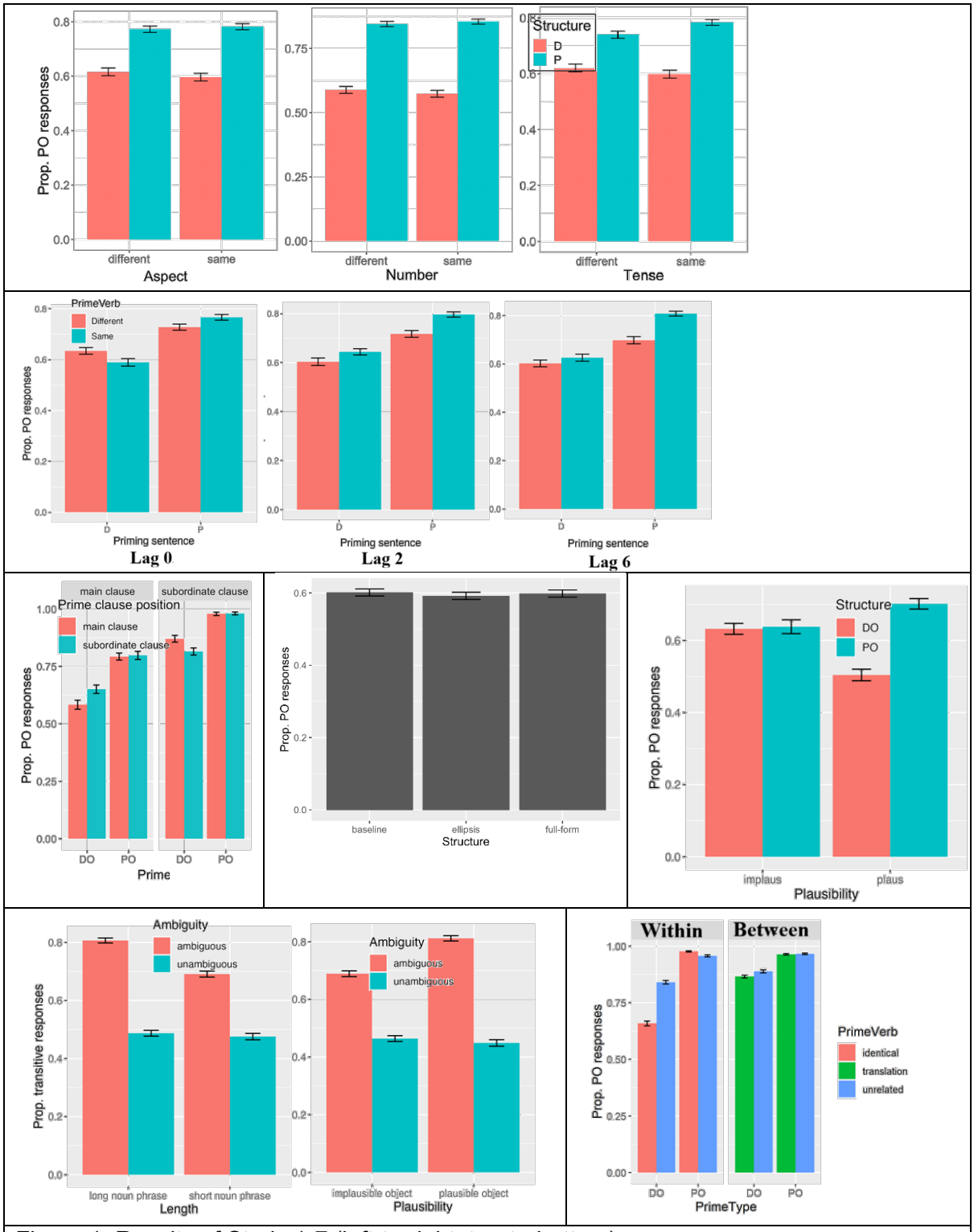

Figure 1: Results of Study 1-7 (left-to-right, top-to-bottom).

References

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